WOLLO UNIVERSITY

KOMBOLCHA INSTITUTE OF TECHNOLOGY

COLLEGE OF INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

Program	**Information Technology**				
Course Code	**ITec**4111				
Course Title	Network Design				
Module Name	Network design, configuration and administration				
Module Number	11				
Instructor's Info	**Name**: Gashaw W. **Cellphone No**: 0913094697/0998064344 **E-mail**: getgashaw4@gmail.com				
Schedule	**Sec_A**: **Wed** 7:40-10:10 **LT and Fr** 2:10-4:00 **LT** **Sec_B**: **Mon** 7:40-10:10 **LT and Wed** 4:00-5:45 **LT**				
Contact Hours	**Lecture**	**Lab/Practical**	**Tutorial**	**Home Study**	**Total**
	2	3	0	5	10
Target Group	4th Year Information Technology Students				
Year /Semester	Year: IV, Semester: I				
Pre-requisites	ITEC2102 (Data Communication and Computer Networks)				
Status of the Course	**Core**				
Course description	This course is intended to teach students how to design and implement computer networks. The course covers detailed networking concepts like transmission media installation, switch and router selection and configuration, connecting to the internet, creating wired and wireless networks, implementing sub netting techniques and others. Students should be equipped with the latest networking technologies like WiFi and how to design an efficient computer network.				

Course Objective	At the end of this course, students will be able to: ❖ Understanding how networks are integrated into business practices; ❖ Design computer networks for enterprises ❖ Deploy a small to medium sized networks ❖ Understanding network simulation principles and methods; ❖ Mastering the systems approach in network design; ❖ Developing practical skills for network design in a heterogeneous environment.
Course Outline	**Chapter 1: Introduction** 1.1. What is network design 1.2. Network design principles **Chapter 2: Switching Design** 2.1. Switching Types 2.2.Spanning Tree Protocol 2.3.Virtual LANs 2.4.Multilayer Switching and Cisco Express Forwarding 2.5.Switching Security 2.6.Switching Design Considerations **Chapter 3: IPv4 Routing Design** 3.1. IPv4 Address Design 3.2.IPv4 Routing Protocols 3.3.IPv4 Routing Protocol Selection 3.4.Design Remote Connectivity **Chapter 4: WAN Design** 4.1. Centralized and Distributed Network Design 4.2.Star and Tree Topology Networks Backbone Networks 4.3.Mesh Networks 4.4.Network Security Design 4.5.Designing Basic Campus and Data Center Networks

	Chapter 5: Wireless LAN Design 5.1. LAN- Design 5.2.Wireless Technology Overview 5.3.Wireless Security 5.4.Wireless Management 5.5.Wireless Design Considerations
Lab Contents	**Design Basic Campus Network** ❖ Design remote network connectivity ❖ Design IP addressing network ❖ Design wireless network
Assessment	❖ Quizzes(x2): 5% ❖ Lab Exam: 15% ❖ Project: 20% ❖ Mid Exam: 20% ❖ Final written exam: 40%
Reference	❖ Teare, D., & Paquet, C. (2006). Campus Network Design Fundamentals. Cisco Press. ❖ Callaway, E. H. (2004). Wireless Sensor Networks: Architectures and protocols. Auerbach Publications. ❖ Doherty, J., Anderson, N., & L., D. M. P. (2008). Cisco Networking Simplified. Cisco Press. ❖ Thomson Learning. (2000). Network design.